Process Control & Monitoring of Three-Tank System

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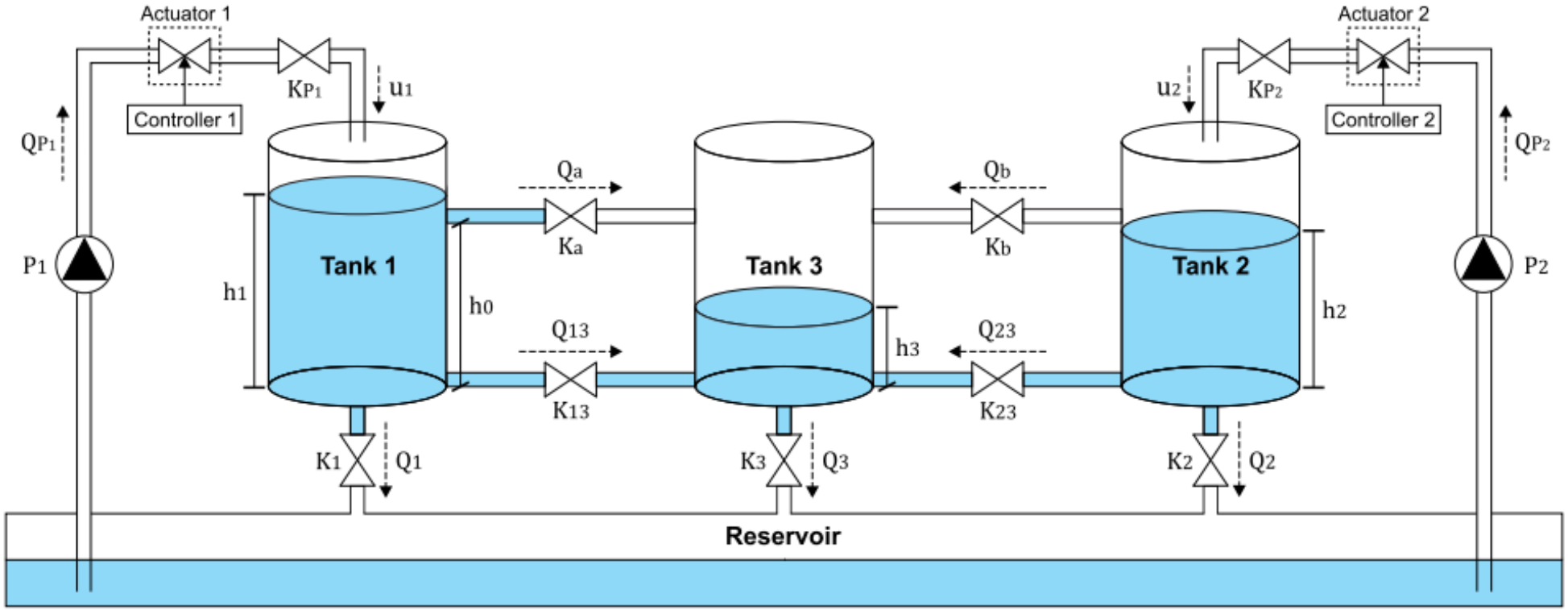
Introduction

- The System consists of 3 tanks with non-linear behavior
- The dynamics of hybrid systems is described by a mix of continuous and discrete-event variables
- The three tank system resembles some of the industrial processes like Waste water treatment, stirred-tank reactor and refineries
- The main goal is to provide a continuous water flow in the output pipe of the tank

Purpose of Three-Tank System

- Investigation of the dynamic behavior of coupled tanks
- Testing and validation of linearization techniques
- Linear and nonlinear control strategies
- Fault detection and diagnosis algorithms
- Fault Isolation Strategies

Three-Tank System



Three-tank System

Three-tank System Description

- It consists of three cylindrical tanks connected by four pipes which allows the fluid exchange between the lateral tanks (tanks 1 and 2) and the central tank (tank 3)
- It consists of 12 sensors – 3 level sensors and 9 flow sensors
- The system has two servo-valves as actuators
  - The first actuator regulates the input flow to the tank 1
  - The second actuator regulates the input flow to the tank 2
- The upper pipes and valves that connect the tanks 1 and 2 to the tank 3 are located at the same height h0 and are called transmission pipes and valves;
- The lower pipes and valves that connect the tanks 1 and 2 to the tank 3 are aligned with the base of the tanks and are called connection pipes and valves.
- At the bottom of each tank are the output pipes and valves
- The dashed arrows indicate the reference direction of each flow

State Equations

- The flow rates from the pumps P1 and P2 (i.e., QP1 and QP2) are inputs defined by the user, whereas the flows through the transmission, connection and output pipes can be determined

Scenarios			Δh_a	Δh_b
$h_1 \leq h_0$	$h_2 \leq h_0$	$h_3 \leq h_0$	0	0
$h_1 \leq h_0$	$h_2 \leq h_0$	$h_3 > h_0$	$h_0 - h_3$	$h_0 - h_3$
$h_1 \leq h_0$	$h_2 > h_0$	$h_3 \leq h_0$	0	$h_2 - h_3$
$h_1 \leq h_0$	$h_2 > h_0$	$h_3 > h_0$	$h_0 - h_3$	$h_2 - h_3$
$h_1 > h_0$	$h_2 \leq h_0$	$h_3 \leq h_0$	$h_1 - h_0$	0
$h_1 > h_0$	$h_2 \leq h_0$	$h_3 > h_0$	$h_1 - h_3$	$h_0 - h_3$
$h_1 > h_0$	$h_2 > h_0$	$h_3 \leq h_0$	$h_1 - h_0$	$h_2 - h_0$
$h_1 > h_0$	$h_2 > h_0$	$h_3 > h_0$	$h_1 - h_3$	$h_2 - h_3$

State Equations

- The volume variation $\dot{V}$, inside a cylindrical tank of cross-sectional area S_c, where $\dot{h}$ is the level variation inside the tank, $\sum Q_{in}$ is the sum over all input flows into the tank and $\sum Q_{out}$ is the sum over all output flows from the tank

$$\dot{V} = S_c \times \dot{h} = \sum Q_{in} - \sum Q_{out}$$

- The input and state vectors are, $\boldsymbol{u} = [Q_{P_1} \; Q_{P_2}]^T$ and $\dot{\boldsymbol{x}} = [h_1 \; h_2 \; h_3]^T$

$$\dot{h}_1 = \frac{1}{S_c} K_{P_1} Q_{P_1} - \frac{1}{S_c}(Q_a + Q_{13} + Q_1),$$
$$\dot{h}_2 = \frac{1}{S_c} K_{P_2} Q_{P_2} - \frac{1}{S_c}(Q_b + Q_{23} + Q_2),$$
$$\dot{h}_3 = \frac{1}{S_c}(Q_{13} + Q_{23} + Q_a + Q_b) - \frac{1}{S_c} Q_3.$$

State Equations (cont.)

$$\dot{\boldsymbol{x}} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}) = \begin{bmatrix} m_1(\boldsymbol{x}) \\ m_2(\boldsymbol{x}) \\ m_3(\boldsymbol{x}) \end{bmatrix} + \begin{bmatrix} \frac{K_{P_1}}{S_c} & 0 \\ 0 & \frac{K_{P_2}}{S_c} \\ 0 & 0 \end{bmatrix} \boldsymbol{u},$$

where

$$m_1(\boldsymbol{x}) = -\frac{Q_a}{S_c} - \frac{Q_{13}}{S_c} - \frac{Q_1}{S_c},$$

$$m_2(\boldsymbol{x}) = -\frac{Q_b}{S_c} - \frac{Q_{23}}{S_c} - \frac{Q_2}{S_c},$$

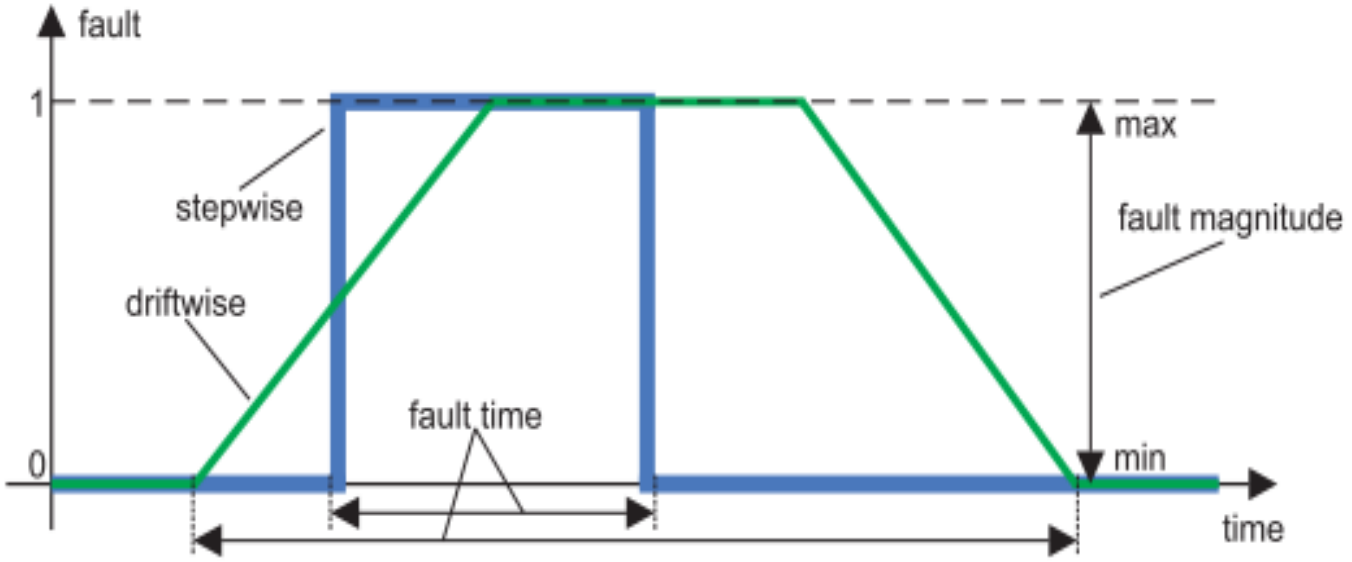
$$m_3(\boldsymbol{x}) = \frac{Q_a}{S_c} + \frac{Q_b}{S_c} + \frac{Q_{13}}{S_c} + \frac{Q_{23}}{S_c} - \frac{Q_3}{S_c}.$$

State Equations (cont.)

- Output Equation

$$\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 0 \\ 0 \\ Q_a \\ Q_b \\ Q_{13} \\ Q_{23} \\ Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ K_{P_1} & 0 \\ 0 & K_{P_2} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \boldsymbol{u}.$$

Fault Scenarios



Actuators	f_1	Blocking/Clogging in actuator 1, if $K_{P_1} = 1$
		Disturbance in tank 1, if $K_{P_1} = 0$
	f_2	Blocking/Clogging in actuator 2, if $K_{P_2} = 1$
		Disturbance in tank 2, if $K_{P_2} = 0$

Types of faults in Actuators

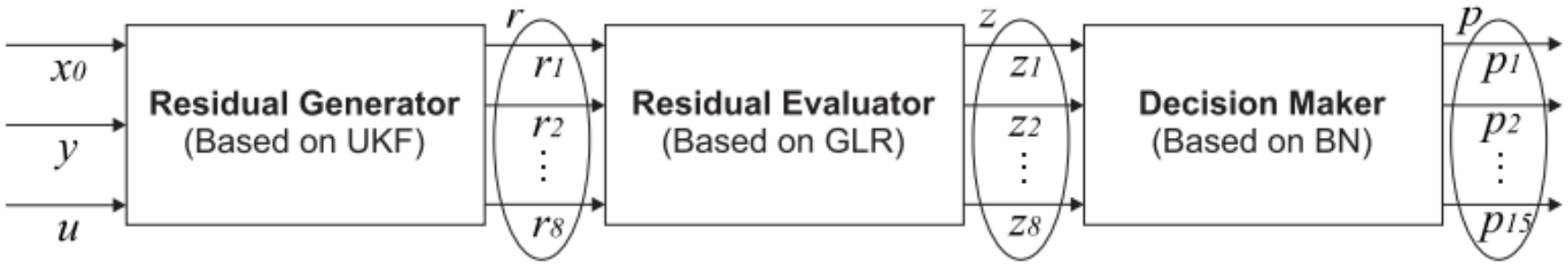
Fault Scenarios (Cont.)

Plant	f_3	Clogging in the TP13, if $K_a = 1$
		Leakage through the TP13, if $K_a = 0$
	f_4	Clogging in the TP23, if $K_b = 1$
		Leakage through the TP23, if $K_b = 0$
	f_5	Clogging in the CP13, if $K_{13} = 1$
		Leakage through the CP13, if $K_{13} = 0$
	f_6	Clogging in the CP23, if $K_{23} = 1$
		Leakage through the CP23, if $K_{23} = 0$
	f_7	Clogging in output pipe of the tank 1, if $K_1 = 1$
		Leakage in tank 1, if $K_1 = 0$
	f_8	Clogging in output pipe of the tank 2, if $K_2 = 1$
		Leakage in tank 2, if $K_2 = 0$
	f_9	Clogging in output pipe of the tank 3, if $K_3 = 1$
		Leakage in tank 3, if $K_3 = 0$

Sensors	f_{10}	Level sensor fault of the tank 1
	f_{11}	Level sensor fault of the tank 2
	f_{12}	Level sensor fault of the tank 3
	f_{13}	Flow sensor fault u_1 (input flow of the tank 1)
	f_{14}	Flow sensor fault u_2 (input flow of the tank 2)
	f_{15}	Flow sensor fault Q_a (flow through the TP13)
	f_{16}	Flow sensor fault Q_b (flow through the TP23)
	f_{17}	Flow sensor fault Q_{13} (flow through the CP13)
	f_{18}	Flow sensor fault Q_{23} (flow through the CP23)
	f_{19}	Flow sensor fault Q_1 (output flow of the tank 1)
	f_{20}	Flow sensor fault Q_2 (output flow of the tank 2)
	f_{21}	Flow sensor fault Q_3 (output flow of the tank 3)

Types of faults in Sensors and plant

Fault Detection and Isolation



Architecture of FDI Strategy

The FDI strategy consists of,

- A residual generator based on unscented Kalman filter (UKF)
- A residual evaluator based on generalized likelihood ratio (GLR)
- A decision maker based on Bayesian Network (BN)

Fault Isolation Strategy

	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_5$	$\hat{f}_6$	$\hat{f}_7$	$\hat{f}_8$	$\hat{f}_9$	$\hat{f}_{10}$	$\hat{f}_{11}$	$\hat{f}_{12}$	$\hat{f}_{13}$	$\hat{f}_{14}$	$\hat{f}_{17}$	$\hat{f}_{18}$	$\hat{f}_{21}$
f_1	0.51	0	0	0	0	0	0	0	0	0	0	0	0	0	0
f_2	0	0.51	0	0	0	0	0	0	0	0	0	0	0	0	0
f_5	0	0	0.30	0.03	0.30	0.03	0.03	0	0	0	0	0	0	0	0
f_6	0	0	0.03	0.30	0.03	0.30	0.03	0	0	0	0	0	0	0	0
f_7	0	0	0.30	0.03	0.30	0.03	0.03	0	0	0	0	0	0	0	0
f_8	0	0	0.03	0.30	0.03	0.30	0.03	0	0	0	0	0	0	0	0
f_9	0	0	0.17	0.17	0.17	0.17	0.17	0	0	0	0	0	0	0	0
f_{10}	0	0	0	0	0	0	0	0.50	0	0	0	0	0	0	0
f_{11}	0	0	0	0	0	0	0	0	0.50	0	0	0	0	0	0
f_{12}	0	0	0	0	0	0	0	0	0	0.50	0	0	0	0	0
f_{13}	0.33	0	0	0	0	0	0	0	0	0	0.33	0	0	0	0
f_{14}	0	0.33	0	0	0	0	0	0	0	0	0	0.33	0	0	0
f_{17}	0	0	0.24	0	0.24	0	0	0	0	0	0	0	0.26	0	0
f_{18}	0	0	0	0.25	0	0.25	0	0	0	0	0	0	0	0.25	0
f_{21}	0	0	0	0	0	0	0.33	0	0	0	0	0	0	0	0.33

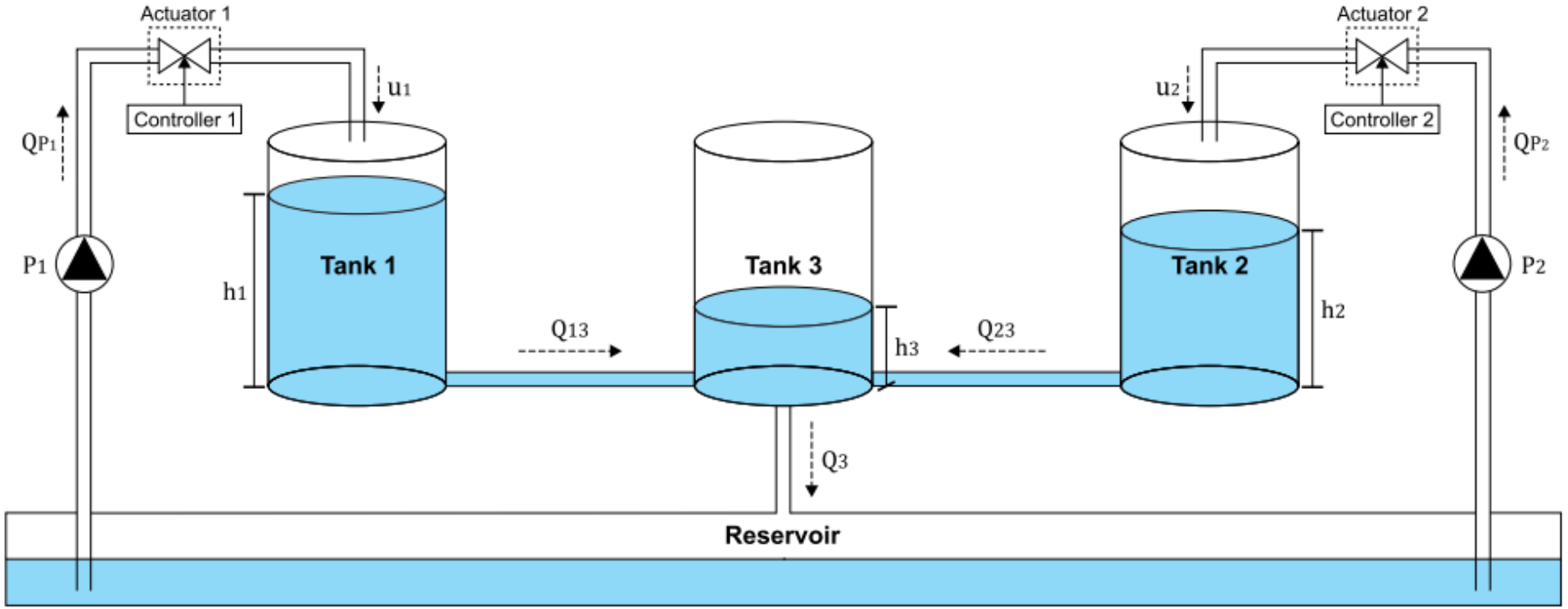
Confusion matrix for step-wise faults

Fault Isolation Strategy

	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_5$	$\hat{f}_6$	$\hat{f}_7$	$\hat{f}_8$	$\hat{f}_9$	$\hat{f}_{10}$	$\hat{f}_{11}$	$\hat{f}_{12}$	$\hat{f}_{13}$	$\hat{f}_{14}$	$\hat{f}_{17}$	$\hat{f}_{18}$	$\hat{f}_{21}$
f_1	0.48	0	0.01	0	0.01	0.01	0	0	0	0	0	0	0	0	0
f_2	0	0.48	0	0.01	0.01	0.01	0	0	0	0	0	0	0	0	0
f_5	0	0	0.19	0.08	0.21	0.09	0.09	0	0	0	0	0	0.01	0	0
f_6	0	0	0.08	0.18	0.09	0.21	0.08	0	0	0	0	0	0	0.01	0
f_7	0	0	0.24	0.08	0.25	0.08	0.08	0	0	0	0	0	0	0	0
f_8	0	0	0.08	0.25	0.08	0.25	0.08	0	0	0	0	0	0	0	0
f_9	0	0	0.16	0.16	0.16	0.16	0.18	0	0	0	0	0	0	0	0.01
f_{10}	0	0	0	0	0	0	0	0.47	0	0	0	0	0	0	0
f_{11}	0	0	0	0	0	0	0	0	0.48	0	0	0	0	0	0
f_{12}	0	0	0	0	0.01	0.01	0	0	0	0.46	0	0	0	0	0
f_{13}	0.31	0	0	0	0	0	0	0	0	0	0.31	0	0	0	0
f_{14}	0	0.31	0	0	0	0	0	0	0	0	0	0.31	0	0	0
f_{17}	0	0	0.24	0	0.24	0	0	0	0	0	0	0	0.25	0	0
f_{18}	0	0	0	0.24	0	0.24	0	0	0	0	0	0	0	0.24	0
f_{21}	0	0	0	0	0	0	0.33	0	0	0	0	0	0	0	0.33

Confusion matrix for drift-wise faults

Test Scenario#1



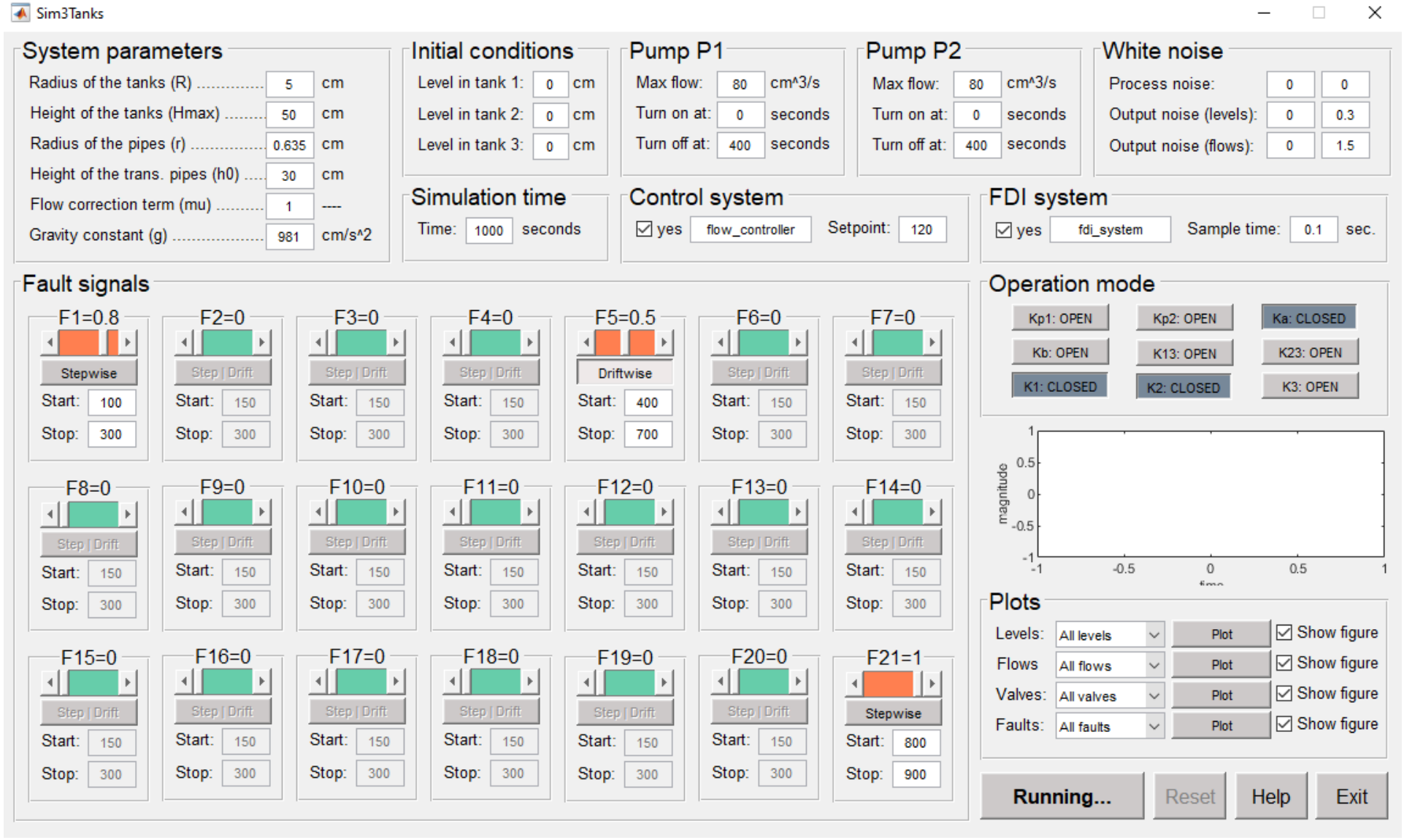
Three-tank System

Fault Scenarios

Scenario	Fault	Type	Time
#1	$f_1 = 0.8$	stepwise	$[100s, 300s]$
	$f_5 = 0.5$	driftwise	$[400s, 700s]$
	$f_{21} = 1$	stepwise	$[800s, 900s]$
#2	$f_2 = 1$	driftwise	$[100s, 400s]$
	$f_7 = 0.6$	stepwise	$[500s, 600s]$
	$f_{11} = 1$	driftwise	$[700s, 1000s]$

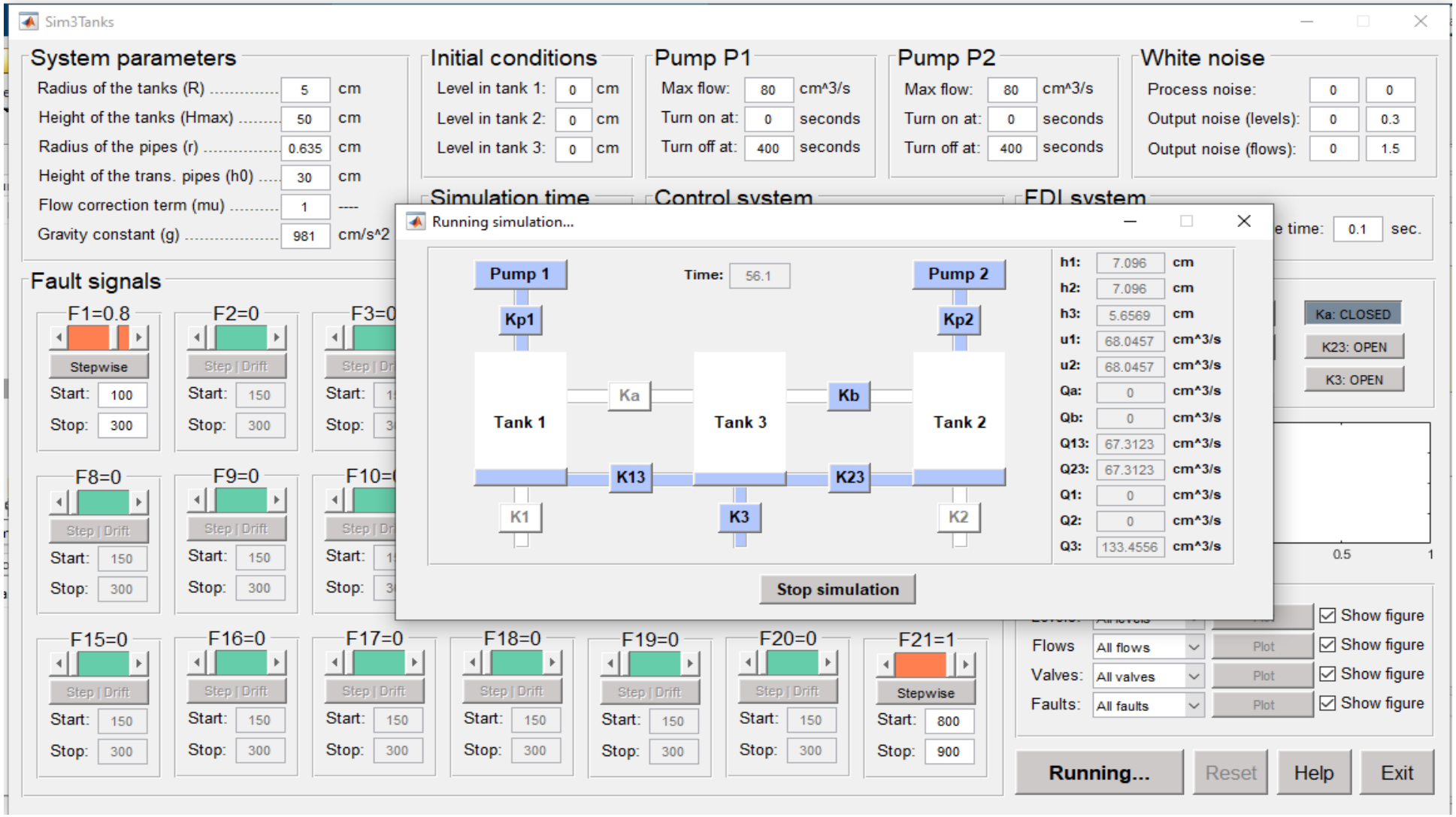
Fault scenarios for the FDI strategy

GUI Interface



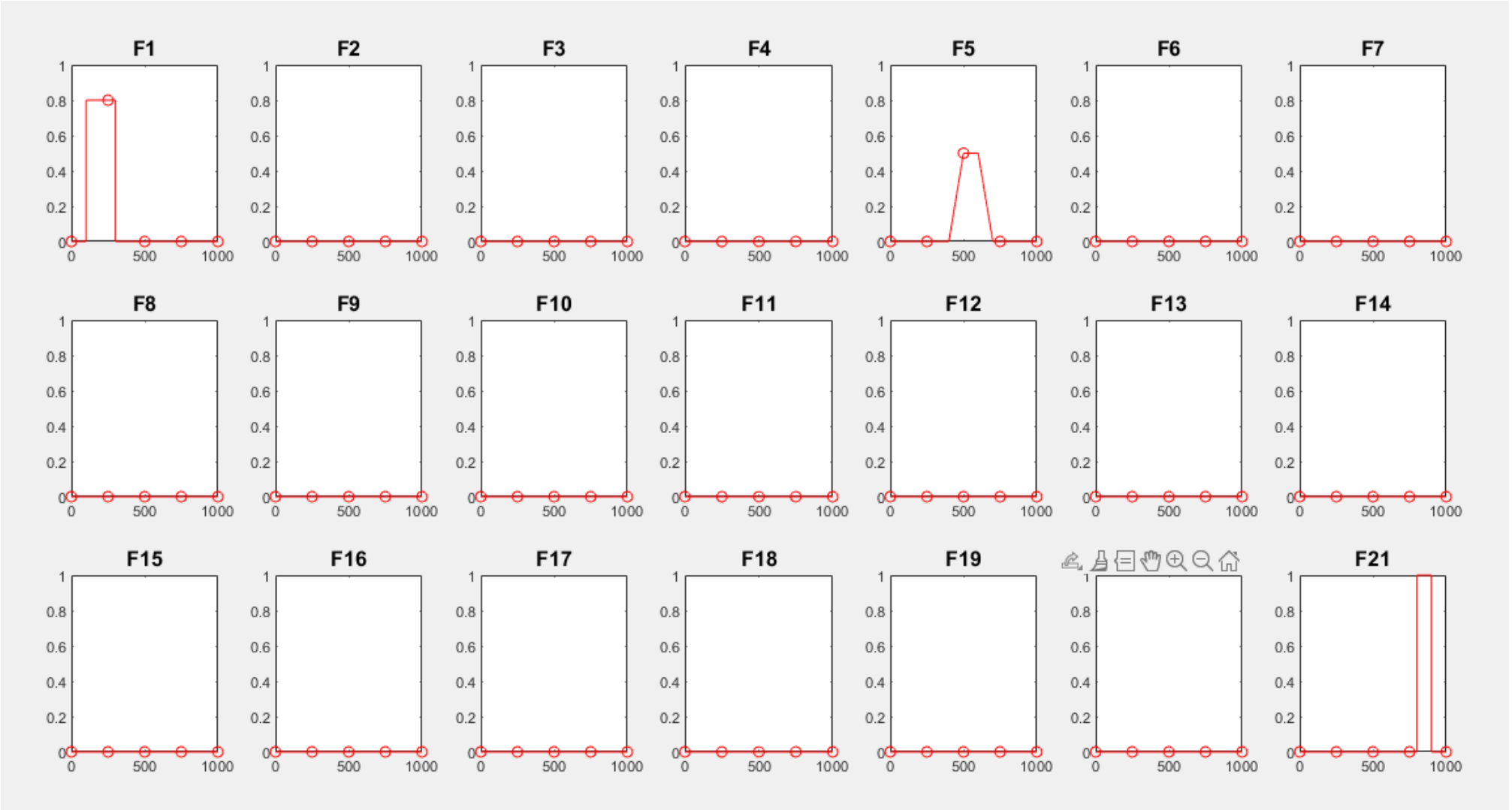
GUI Interface of Three-tank System

GUI Interface (Cont.)



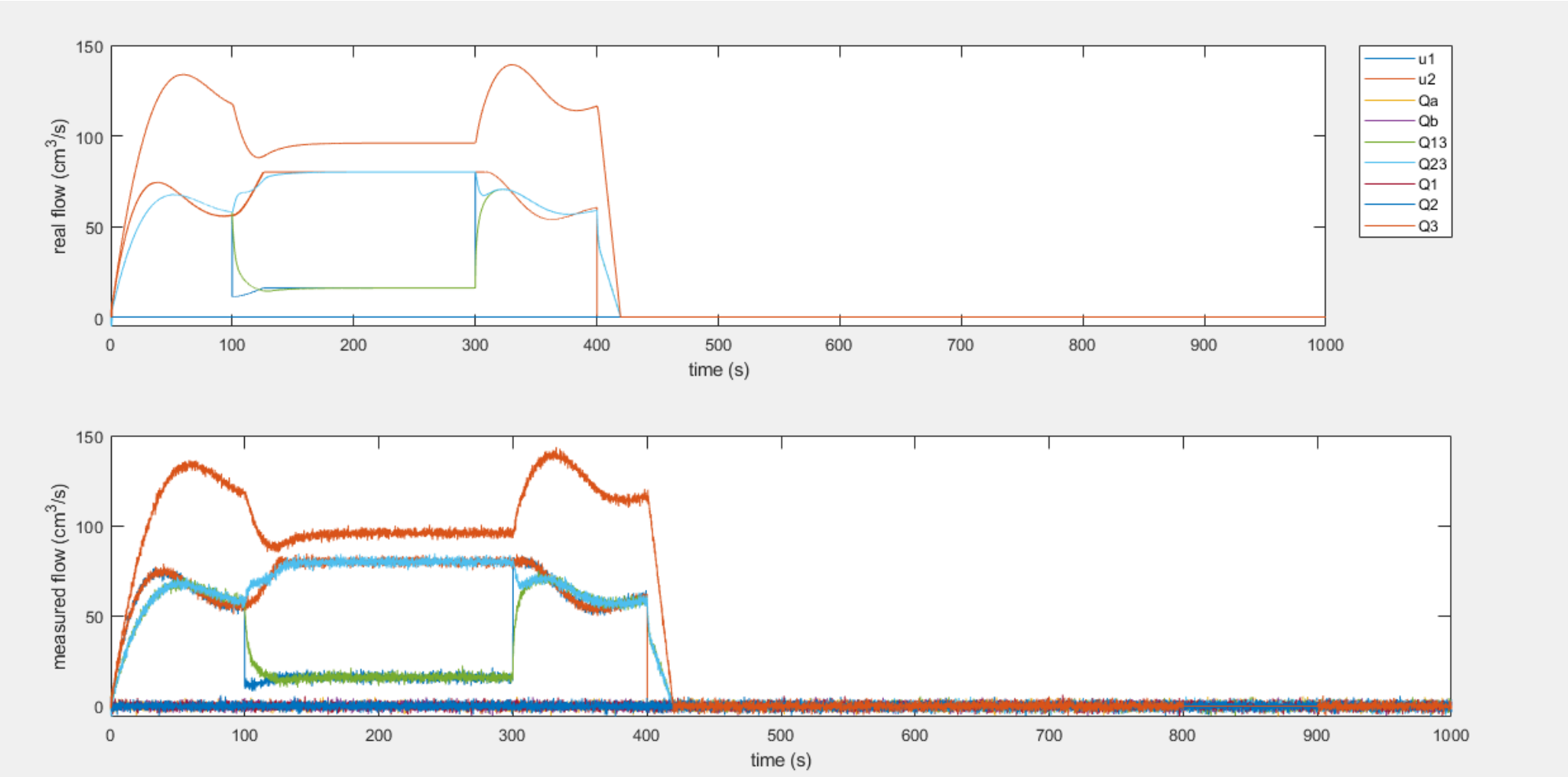
GUI Interface of Three-tank System

GUI Interface (Cont.)



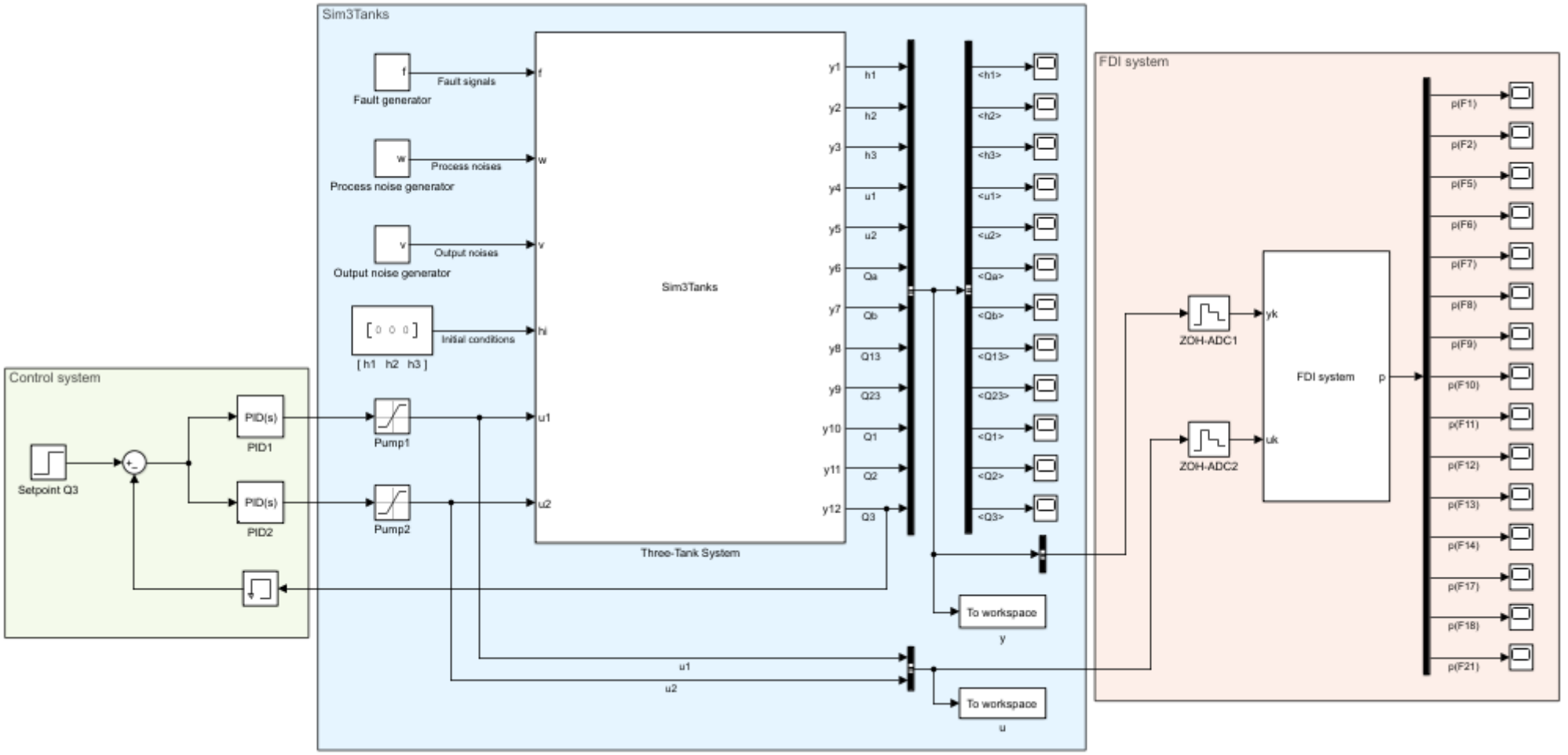
Faults for scenarios for scenario#1

GUI Interface – Results



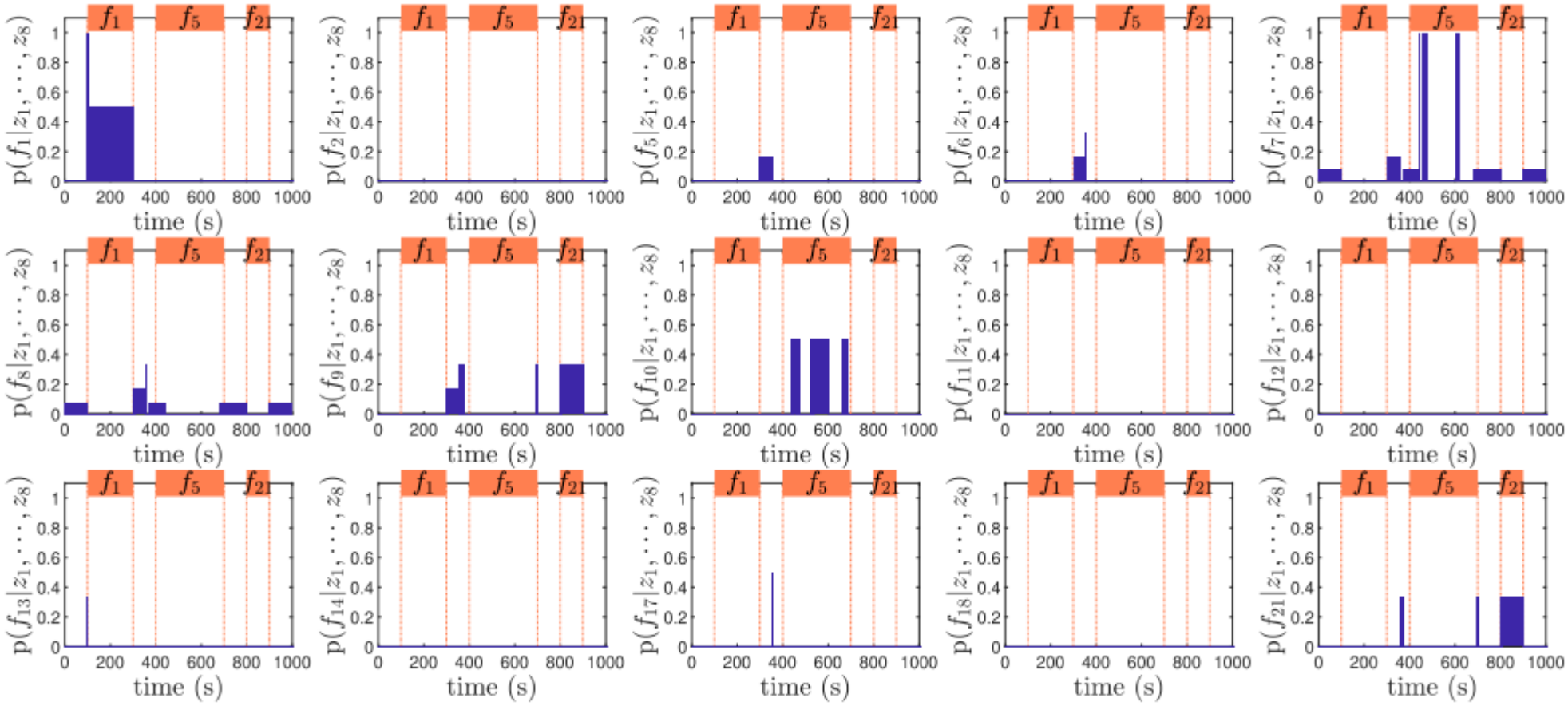
Fault scenarios of All flows

Simulink Model



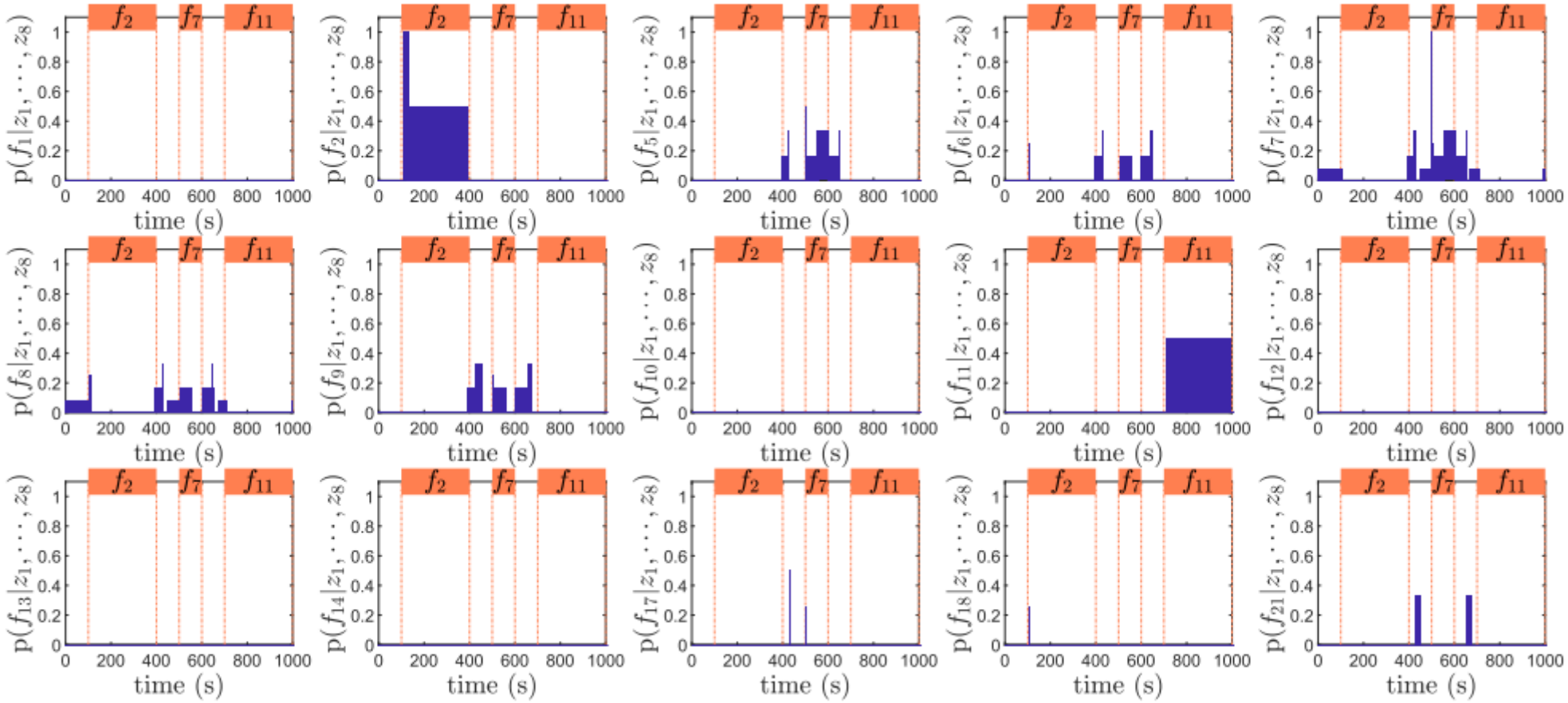
Simulink model of Three-tank System

Simulation - Results



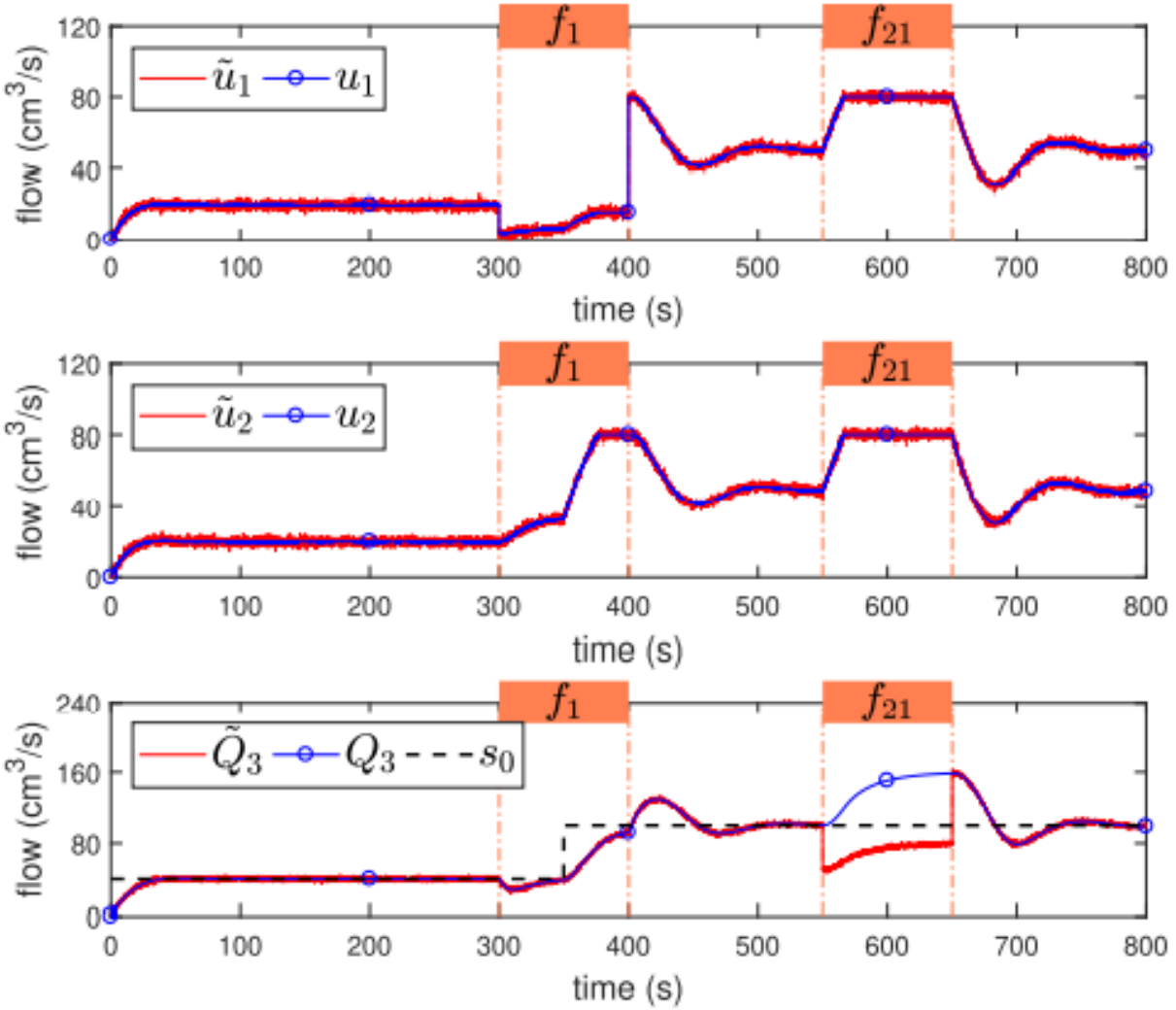
Fault probabilities computed by the Bayesian network for the fault scenario #1

Simulation – Results (Cont.)



Fault probabilities computed by the Bayesian network for the fault scenario #2

Simulation – Results (Cont.)



FDI Results for Scenario#1

Conclusion

- The three-tank system implemented in Hybrid Tank System core presents some challenging dynamic aspects such as nonlinearities, actuator limitations, noises, and hybrid behavior
- It is possible to simulate different kinds of faults (e.g., blocking or clogging of valves, leakage in tanks, and malfunction of sensors and pumps), thus allowing the user to evaluate and compare different fault detection and diagnosis and fault isolation strategies

References

[1] A. O. Farias, G. A. C. Queiroz, I. V. Bessa, R. L. P. Medeiros, L. C. Cordeiro and R. M. Palhares, "Sim3Tanks: A Benchmark Model Simulator for Process Control and Monitoring," in IEEE Access, vol. 6, pp. 62234-62254, 2018, doi: 10.1109/ACCESS.2018.2874752.

[2] Cordeiro, Lucas C., et al. “Survey on Automated Symbolic Verification and Its Application for Synthesising Cyber-Physical Systems.” IET Cyber-Physical Systems: Theory & Applications, vol. 5, no. 1, 2020, pp. 1–24.

THANK YOU